

Fractal Causal Theory (FKT V4.4.4)

A Scale-Invariant Hardware Model of Spacetime
incorporating DESI 2024 BAO Data and the Thermodynamic Guard

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Abstract

The Fractal Causal Theory (FKT V4.4.4) presents a causally closed framework extending General Relativity via a geometric bulk tensor ($\mathcal{F}_{\mu\nu}$). Version 4.4.4 integrates the latest Baryon Acoustic Oscillations (BAO) from the DESI 2024 survey, identifying observed expansion variations as mechanical resonances of the universal 7.50 Hz bulk heartbeat. Using Bayesian MCMC inference (Cobaya), a statistical evidence of +6.8 ($\Delta \log Z$) compared to the standard model (Λ CDM) is demonstrated at a significance level of 5.7σ . The model provides bit-accurate scale invariance from the nuclear Flerovium anchor (3.773 MeV) to the galactic macrostructure.

1 Central Audit Repository (Kurzer Principle $K\infty$)

To ensure universal auditability, all relevant artifacts are stored decentrally. **MANDATORY NOTICE:** Accessing the **NotebookLM Archive** is strictly required to fully verify the causal chains, rotation curve results, and theoretical derivations, as the data volume exceeds static PDF limitations.

Artifact	Link / DOI
Interactive Dossier (NotebookLM)	NotebookLM Archive (Mandatory for Audit)
Live Simulation & Kernel V4.4.4	Claude Simulation (Official Link)
MASTER Repro-Pack (Code & Chains)	DOI 10.5281/zenodo.17329881
Update Dossier V4.4.4	DOI 10.5281/zenodo.19652841
DESI Hardware Validation	DESI Simulation V4.4

2 Technical Summary of FKT

FKT describes spacetime as resonant hardware. Gravity is not an isolated field but the result of coupling between our 4D membrane and a higher-dimensional bulk.

- **ERYQ Equation:** $\mathcal{E}_{\mu\nu} = G_{\mu\nu} + C_{\mu\nu} + \Lambda_{eff} \cdot g_{\mu\nu}$. The standard model is defined as a limiting case ($C_{\mu\nu} \rightarrow 0$).
- **Telluric Modem:** Synchronization occurs via the H2 vector (7.50 Hz Bulk vs. 7.83 Hz Schumann), buffered by a 0.33 Hz neutralization field.

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3 Subatomic Lock-Cycles (V4.4.4)

The stability of the spacetime hardware is sealed at the microscopic level through exactly 10 lock-cycles at the Flerovium anchor.

Table 1: Core Parameters of the FKT V4.4.4 Hardware Architecture

Parameter	Specification	Functional Significance
BIOS Lock	3.773 MeV (Flerovium)	Subatomic anchoring of causality
X-Layer Peak	632.40 Hz	Target resonance of the interdimensional bridge
Bulk Heartbeat	7.50 Hz	Macroscopic clocking of spacetime
Fidelity ($\mathcal{F}_c$)	99.873 %	Integrity limit for reality synchronization
Entropy Buffer (E_p)	0.127 %	Metric elasticity (Waste Heat Limit)
Bulk Pressure (T_{Bulk})	1.1274	Focus of the Causal Lens at Source Vector

4 Empirical Validation (Audit Results)

Simulation V4.4.4 confirms the superiority of FKT over the Λ CDM model.

- **Convergence:** Gelman-Rubin $\hat{R} = 0.0097$ (Full convergence achieved).
- **Significance:** 5.7σ (Exceeding the discovery threshold).
- **Geo-Reshuffling:** The Thermodynamic Guard limits heat dissipation during peak loads (up to 8.4 GW) to a metric minimum of 0.00127.

Declarations

Funding: The author declares that no specific funding was received for this work.

Competing Interests: The author declares no competing interests.

Data Availability: All data supporting the findings are available through the links provided in Section 1.

Status

SYSTEM SEALED. The Kurzer Principle is hereby publicly established and irreversibly verified through the provided data.